

TECHNICAL AND EPIDEMIOLOGICAL REFERENCE DOSSIER

An Edge-Native, Multi-Modal Early-Warning Platform for Subclinical Bovine Mastitis in Indian Dairy Herds

Team Name: H14_SPIRIT

Team ID: 146962

Problem Statement Number: 26109

Journey stage: Understood completely

Section 1 — Executive Summary and Epidemiological Problem Space

1.1 Purpose of This Dossier

This dossier consolidates the clinical, epidemiological, engineering, and economic foundations underlying MastitisGuard AI, an edge-native early-warning platform for subclinical bovine mastitis developed by Team H14_SPIRIT (Team ID 146962), Problem Statement 26109. Every claim in this dossier distinguishes what the current prototype demonstrates today from what remains a target for real-farm validation; the exact same distinction is carried through the Decision-Making, ML Model Documentation, and Validation & Justification documents.

1.2 Clinical Pathology of Bovine Mastitis

Bovine mastitis is an inflammatory response of mammary gland tissue, almost always triggered by intramammary bacterial infection entering through the teat canal. The disease exists on a continuum rather than as a binary state:

- **Subclinical intramammary infection (SCM):** Pathogens colonize the mammary gland and provoke a measurable immune response — elevated somatic cell count (SCC), altered milk electrical properties, and shifting milk pH — without visible signs in the udder or milk. This is the stage at which most economic loss accrues, precisely because it is invisible to the farmer.
- **Progression to clinical flare-up:** As bacterial load and inflammatory mediators increase, the gland develops visible swelling, heat, and pain, and the milk shows clots, flakes, or serous discoloration. This is the stage at which most Indian smallholders first detect the disease, by which point tissue damage is already underway.
- **Alveolar necrosis and fibrosis:** Untreated or repeated clinical episodes destroy secretory alveolar tissue, which is replaced by non-functional fibrous scar tissue. This damage is irreversible.
- **Blind quarters:** In severe or repeated cases, an entire quarter loses secretory function permanently, structurally capping the animal's lifetime milk yield regardless of any subsequent treatment.

The central epidemiological insight motivating MastitisGuard AI is that the subclinical stage precedes clinical presentation by days, and physiological signals — udder-adjacent skin temperature, rumination and activity behaviour, and milk chemistry — begin drifting from an animal's own baseline before a farmer would notice anything by eye. A system that can detect this drift early converts a reactive, damage-already-done event into a manageable, low-cost intervention window.

1.3 The Indian Dairy Landscape

- Over 80 million smallholder households participate in milk production, most operating far below the scale of organized commercial dairies.
- The national milch cattle population exceeds 112 million animals.
- National milk output stands at approximately 247.87 million tonnes annually, making India the world's largest milk producer by volume.
- The typical herd size at the smallholder level is 2 to 5 cows, managed with minimal mechanization and largely manual husbandry practices.

This landscape means any viable technology must be inexpensive per animal, function without dependence on continuous internet connectivity, and integrate into existing manual milking routines rather than requiring parlor-scale automation — the exact gap quantified in the Market Research document.

1.4 Macroeconomic Burden of Mastitis in India

- **National annual loss:** Estimated between ₹7,000 crore and ₹11,500 crore per year across the Indian dairy sector.
- **Share attributable to subclinical disease:** SCM is estimated to drive 58% to 70% of this total economic drain, precisely because it goes undetected long enough to suppress yield and quality without triggering treatment.
- **Prevalence:** Pooled prevalence estimates place subclinical mastitis at approximately 45% across surveyed Indian cattle populations.

1.5 The Reactive Crisis at the Farmer Level

- **Milk yield collapse:** A daily milk drop of 3–4 kg per affected cow, translating to a direct daily revenue loss of approximately ₹306–₹458 per day at prevailing farm-gate milk prices.
- **Mandatory withdrawal-period dumping:** Once antibiotic treatment begins, regulations mandate a 3–7 day withdrawal period during which milk must be discarded, adding a further ₹150–₹200 per day in dumped-milk losses.
- **Out-of-pocket treatment costs:** A single clinical mastitis episode typically costs the household ₹3,500–₹6,000 in veterinary fees, antibiotics, and associated care.

For a smallholder household with 2–5 cows, a single undetected subclinical case that progresses to clinical mastitis can represent several weeks of net dairy income. MastitisGuard AI's current prototype demonstrates a 1–5 day pre-clinical drift window on synthetic, physiologically-grounded data (see ML Model Documentation, Section 5); the 7–14 day figure cited in veterinary literature is the Phase I real-farm validation target this project is designed to reach, not a result already demonstrated — the same distinction is made in the ML Model Documentation and Validation & Justification documents.

Section 2 — Decoupled Three-Unit Hardware Topology

2.1 Why Western Precision-Dairy Systems Fail in the Indian Context

Commercial precision dairy farming (PLF) sensor systems developed for Western markets are built around mechanized milking parlors: inline conductivity and yield sensors sit inside automated milking clusters that pass every litre of milk through fixed measurement hardware. These imported inline parlor sensor systems typically cost ₹20,000–₹40,000 per animal and presuppose infrastructure a 2–5 cow smallholder herd

milked by hand will never have. MastitisGuard AI is instead built around a decoupled, **three-unit topology that separates animal-worn sensing, milking-point sensing, and the compute/aggregation layer**, so each unit can be built from low-cost commodity components and the system degrades gracefully rather than requiring a parlor at all.

2.2 Unit 1 — **Smart Wearable Collar**

Bill of Materials: **₹2,510 per cow**, prototype grade (full itemisation in the Bill of Materials document, Section 2 — identical figure).

2.2.1 **Controller**

- **ESP32 DevKit V1** — dual-core microcontroller with integrated **Wi-Fi** and **Bluetooth**, running FreeRTOS tasks so sensor polling, deep-sleep scheduling, and **ESP-NOW** transmission execute as independent, non-blocking tasks.
- **Deep-sleep power management** — the controller wakes on a scheduled duty cycle to sample sensors and transmit, then returns to deep sleep, the dominant lever for multi-day battery life on the 18650 cell.

2.2.2 **Sensors**

- **MAX30205 clinical-grade contact temperature sensor**: placed at the dewlap for stable skin contact, chosen because it is a medical-accuracy digital temperature IC ($\pm 0.1^\circ\text{C}$ typical) rather than a generic thermistor. Current bench prototype uses a DHT11 as a temporary stand-in for early firmware and mesh testing; the MAX30205 is the specified and costed component for the pilot build (see Bill of Materials, Section 6, Build Status Disclosure).
- **MPU6050 6-axis IMU**: combined **accelerometer** and **gyroscope** used to derive an activity index and a rumination-chewing proxy, established behavioural indicators of illness onset in cattle.
- **NEO-6M GPS module**: tracks pasture **movement** and **grazing range**, complementing the IMU's fine-grained activity signal.

2.2.3 **Enclosure and Power**

- **3D-printed (FDM/SLA), IP65-equivalent enclosure** for the current prototype — rated for outdoor grazing, sun exposure, and incidental water contact. Certified IP69K injection-moulded housing is a Phase 2 production target, not the current build (see Bill of Materials, Section 6).
- **3.7V 2,600 mAh 18650 lithium-ion cell** as the primary power source.
- **TP4056 power-management IC** for single-cell lithium charge management and protection.
- High-tensile woven strap pass-throughs so the collar securely rides the neck through normal grazing and herd movement.

2.3 Unit 2 — **Shared Milking Station Node**

Bill of Materials: **₹2,110 per shed**, prototype grade (shared across the whole herd at a milking point, not per animal — the key cost lever that keeps per-cow economics viable at smallholder scale). A field-grade variant at ₹5,250 is budgeted for Phase 2 scale-up (see Bill of Materials, Section 3, for the full breakdown of that upgrade).

2.3.1 Dual-Controller Architecture

The milking station node runs two coordinated controllers: an ESP32-CAM handles optical animal identification, while a secondary ESP32 DevKit V1 handles the analog and digital instrumentation bus (pH, temperature, load cell, and CMT staging input), so camera-driven image workloads never block time-critical sensor sampling.

2.3.2 Identification

Optical scanning of ear-tag QR codes or barcodes via the OV2640 camera module links each milking session's readings to a specific Cow ID (for example, COW-023), the mechanism that lets the system maintain a distinct rolling physiological baseline per animal rather than a single herd-wide average.

2.3.3 Instrumentation

- **PHSENS analog pH probe (BNC interface):** measures milk pH at the point of milking; normal raw milk pH sits in the 6.5–6.8 range, while mastitis-associated inflammation shifts milk toward an alkaline 6.9–7.4 range. Budgeted and tested in firmware/data structures; physical integration into the milking manifold is the defined Phase 1 pilot task (see Bill of Materials, Section 6).
- **DS18B20 waterproof digital temperature probe:** provides thermal compensation for the pH reading, since probe response is temperature-dependent.
- **HX711 24-bit ADC with 40 kg load cell:** measures per-session milk yield, so yield suppression at the individual-cow, individual-session level becomes a quantitative input rather than an anecdotal farmer observation.
- **Manual California Mastitis Test (CMT) staging paddle:** provides a low-cost, on-the-spot confirmatory test used for immediate farmer decision support and for held-out model validation — never as a model input feature (see ML Model Documentation, Section 5, for why).

2.4 Unit 3 — Central Shed ESP Hub (Local Edge Gateway)

Bill of Materials: ₹950 per shed (full itemisation in the Bill of Materials document — identical figure).

2.4.1 Hardware Role

The shed hub is the local aggregator and compute engine: it receives per-cow, per-session telemetry from the collar and milking-station units, runs the rolling-baseline computation and the CatBoost risk-scoring model described in the ML Model Documentation, and holds the results for farmer and app/dashboard consumption.

2.4.2 Networking

The system is 100% offline by design at the per-cow level: collar-to-hub and milking-station-to-hub communication runs over a peer-to-peer 2.4 GHz ESP-NOW wireless mesh protocol, which requires no Wi-Fi router, no SIM card, and no cloud round-trip for the core forecasting loop to function. This is a deliberate architectural choice for connectivity-constrained rural sheds, not a fallback mode.

2.4.3 Security and Scope

The hub acts as a farm-isolated data buffer: telemetry and computed risk scores for one farm's herd are not transmitted to or comingled with any other farm's data at the edge layer, preventing cross-herd data contamination, unless and until it is deliberately synced upward (for example, to a cooperative-level dashboard).

Section 3 — Where the Predictive Logic Lives

The Shed ESP Hub runs two layers, both described in full technical detail in the ML Model Documentation:

- A deterministic per-cow rolling-baseline layer (7–14 day moving median and deviation residual) — always active, needs no training data, and gives an interpretable drift signal from day one of deployment.
- A CatBoost risk-classification layer trained on top of that baseline data — currently trained and evaluated on physiologically-grounded synthetic data (ROC-AUC 0.964; see ML Model Documentation, Section 5) and exported to run fully offline on the ESP32. Real-farm field accuracy is the Phase 1 validation milestone, not yet a demonstrated result.